



**Team  
Rutgers**



# **Tunneling Instructions**

# QEpy: Hands-on!

Tunneling Option 1

<https://portal.expense.sdsc.edu/pun/sys/dashboard>

```
Account: csd973
Partition: shared
Time limit (min): 1200
Number of cores: 32
Memory required per node (GB): 64
GPUs (optional): 0
Reservation: QDMS-CPU
Working Directory: /expense/lustre/projects/csd973/your_user_name_here/
```

# QEpy: Hands-on!

## Tunneling Option 1

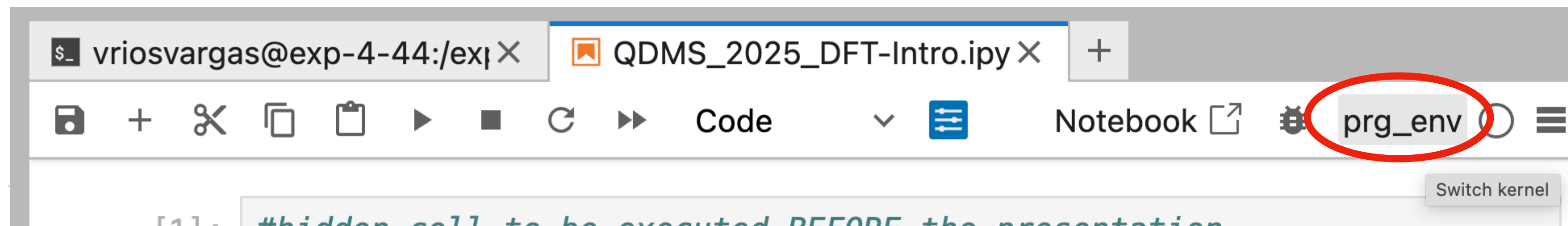
### 1) Open Terminal:

#### a. Activate the Python environment and create a Jupyter kernel

```
$ conda deactivate  
  
$ source /expanse/lustre/projects/csd973/jmartinez2/software/prg_env/bin/activate  
  
$ ipython kernel install --user --name=prg_env  
  
$ cp -r /expanse/lustre/projects/csd973/vriosvargas/DFT-Intro/2025/QDMS/qepy .
```

### 2) Select the kernel in Jupyter:

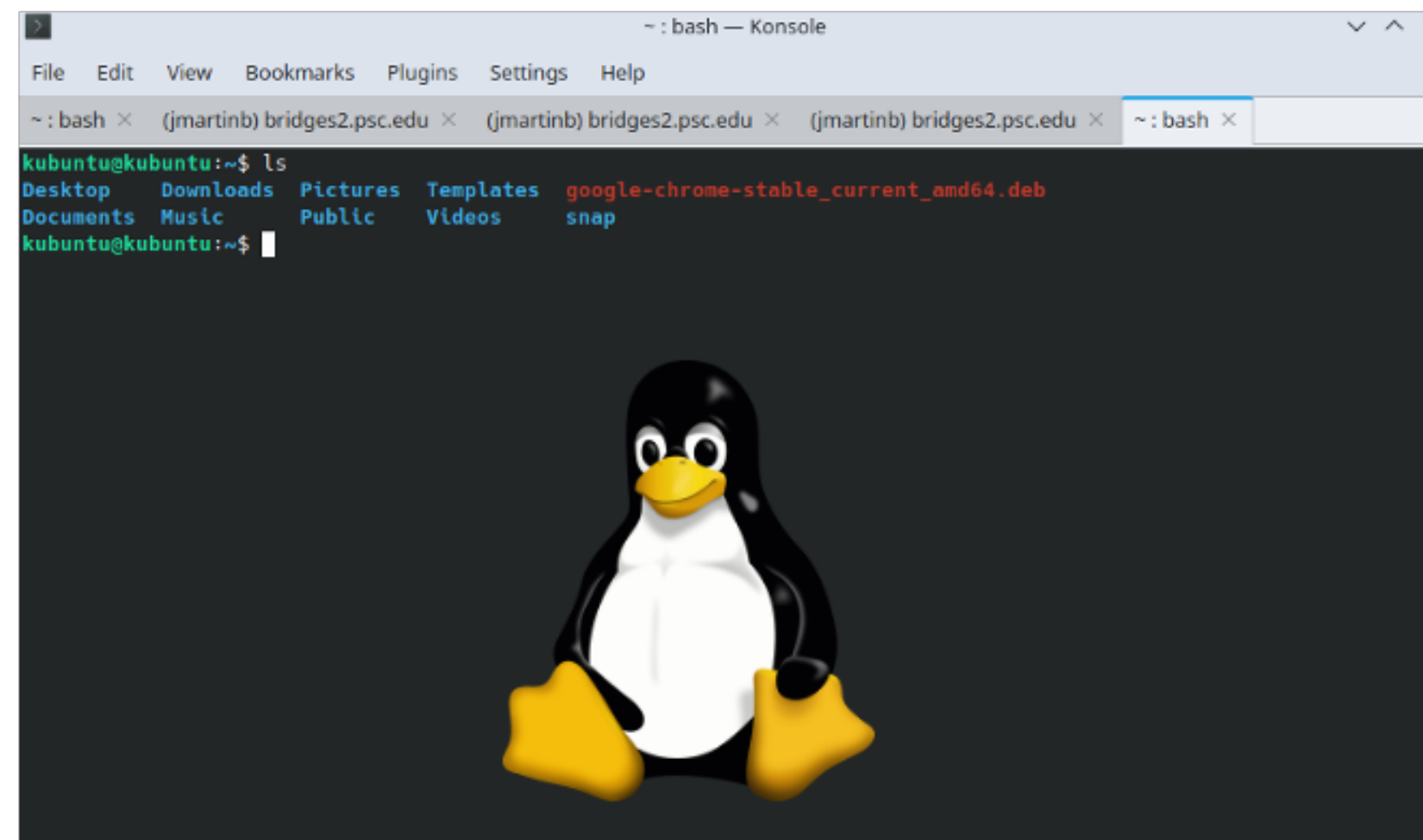
#### a. Activate the Python environment and create a Jupyter kernel



# eDFTpy: Hands-on!

## Tunneling Option 2

### 1) Open Unix Terminal



### 2) ssh to expanse cluster

```
ssh user_name@login.expanse.sdsc.edu
```

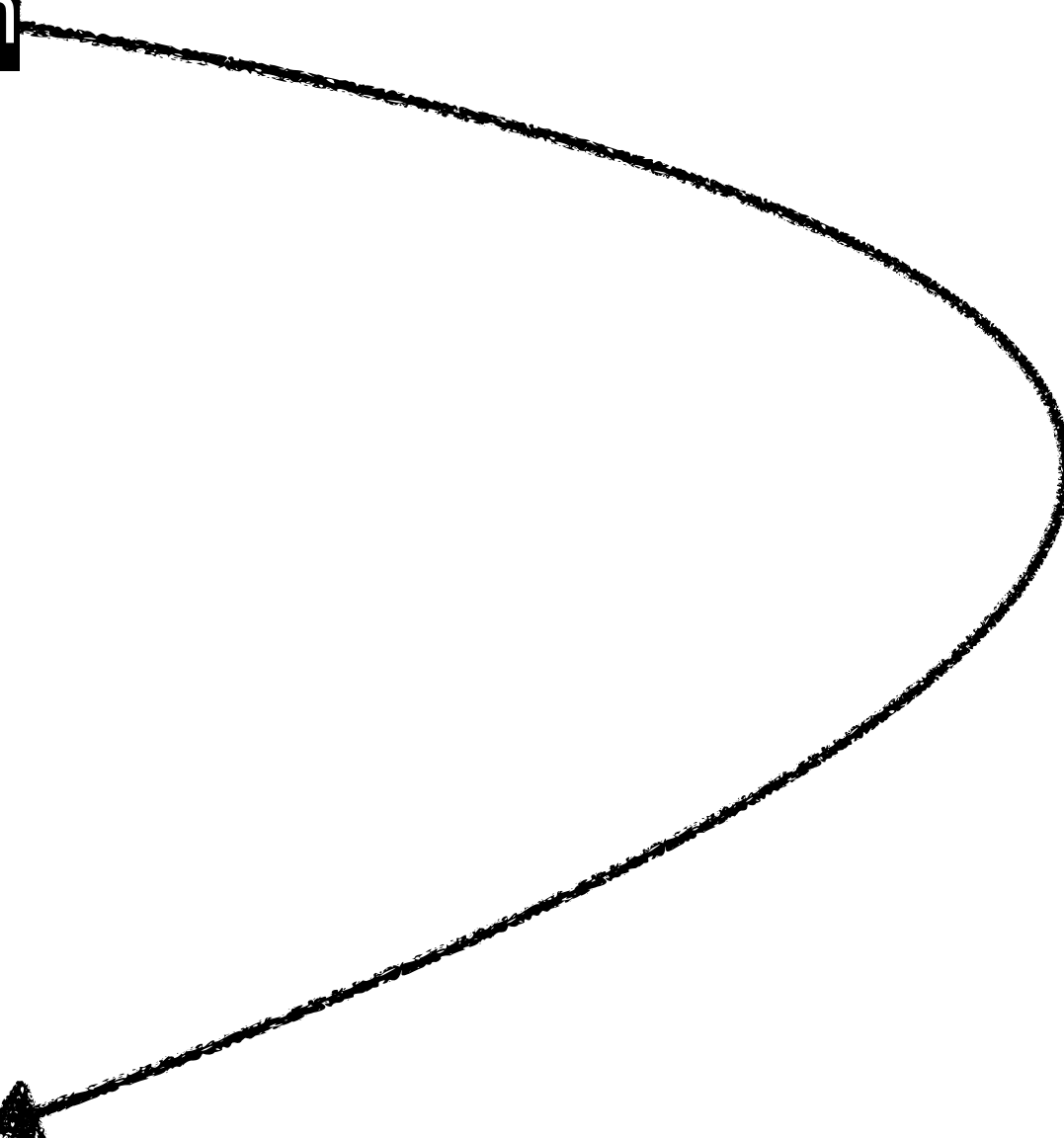
# eDFTpy: Hands-on!

## Tunneling Option 2

Run on compute node

### 3) Create a SLURM file

```
vi run_jupenv.sh
```



### 4) Submit the SLURM file

```
sbatch run_jupenv.sh
```

```
#!/bin/bash
#SBATCH --job-name="tunn"
#SBATCH --output="tunn.%j.%N.out"
#SBATCH --error="tunn.%j.%N.out"
#SBATCH --partition=shared
#SBATCH --reservation=QDMS-CPU
#SBATCH --nodes=1
#SBATCH --ntasks-per-node=20
#SBATCH --mem=0
#SBATCH --account=csd973
#SBATCH --export=ALL
#SBATCH -t 12:00:00
```

```
module purge
module load slurm
module load cpu/0.17.3b gcc/10.2.0/npcyll4
module load python/3.8.12/7zdjza7
module load intel-mkl/2020.4.304/ghfk3mu intel-mpi/
2019.10.317/kdx4qap
```

```
export OMP_NUM_THREADS=12
```

```
env=prg_env
source /expance/lustre/projects/csd973/jmartinez2/
software/$env/bin/activate
ipython kernel install --user --name=prg_env
```

```
echo "Python version: $(python --version)"
echo "Jupyter version: $(jupyter --version)"
echo "Kernel specs:"
jupyter kernelspec list
```

```
jupyter notebook --no-browser --ip=0.0.0.0 --port=PQRS
```

Change the port  
to any 4 digit  
number

# eDFTpy: Hands-on!

## Tunneling Option 2

5) Check that your job is running: `squeue -u user_name`

| JOBID    | PARTITION | NAME     | USER     | ST | TIME    | NODES       | NODELIST(REASON)  |
|----------|-----------|----------|----------|----|---------|-------------|-------------------|
| 41042506 | compute   | tunnelin | jmartine | R  | 4:07:58 | 1 exp-12-33 | ← <b>NODENAME</b> |

6) Search in the `slurm_node.err` file and look for the address that starts with `http` (This will have the port you used in the `sbatch` file): `grep "http\:\:\//127" tunn.*.out | head -1`

7) It will print: `[I 2025-07-10 10:56:03.455 ServerApp] http://127.0.0.1:PQRS/tree?token=ea36626a5564b1ae0c9cf3d4803fe1587fdb717800370b7d` → Keep it!

8) Open a NEW terminal and type: `ssh -L XYZW:NODENAME:PQRS user_name@login.expense.sdsc.edu`

↓  
The **XYZW** is another random number!

# eDFTpy: Hands-on!

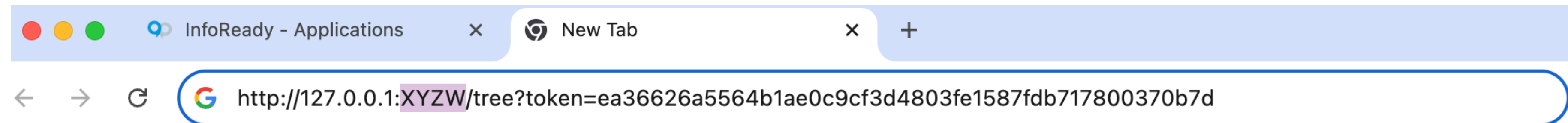
## Tunneling Option 2

9) Copy-and-paste (step 7) to a browser on local computer and change **PQRS** to **XYZW**:

`http://127.0.0.1:PQRS/tree?token=ea36626a5564b1ae0c9cf3d4803fe1587fdb717800370b7d`



`http://127.0.0.1:XYZW/tree?token=ea36626a5564b1ae0c9cf3d4803fe1587fdb717800370b7d`





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**QEpY Tutorial  
folder info**

# QEpy: Hands-on!

Open JN

1) ssh to expanse cluster `ssh user_name@login.expanse.sdsc.edu`

2) pwd to check you're in your working directory

```
/expanse/lustre/projects/csd973/your user name here
```

3) Copy the work directory or git clone:

```
cp -r /expanse/lustre/projects/csd973/vriosvargas/DFT-Intro/2025/QDMS/qepy .
```

If you want to have access to it in the future you can also git clone

```
git clone https://github.com/Quantum-MultiScale-Template/DFT-Intro.git
```

4) Go to the folder and check the JN inside the folder:

```
cd /expanse/lustre/projects/csd973/your user name here/qepy/
```



| / ... / QDMS / qepy /     |                |
|---------------------------|----------------|
| Name                      | Modified       |
| band                      | 29 minutes ago |
| eos                       | 29 minutes ago |
| testing                   | 29 minutes ago |
| QDMS_2025_DFT-Intro.ipynb | 29 minutes ago |